
EFFECTS OF VIDEO MODELING ON TEACHING BIDS FOR JOINT ATTENTION TO CHILDREN WITH AUTISM

Nikki A. Rudy, Alison M. Betz*, Evadne Malone, Justine E. Henry and Ivy M. Chong

The Scott Center for Autism Treatment, Florida Institute of Technology, Melbourne, FL, 32901 USA

The purpose of this study was to evaluate the effects of video modeling to teach children with autism to initiate bids for joint attention. The video model consisted of a child demonstrating three components of a bid for joint attention with an adult conversational partner: orienting toward the object, emitting a vocal statement, and eye gaze shift toward unique objects in the environment. Results indicated that video modeling alone was effective in teaching all components of joint attention for two of the three children, whereas video modeling plus in vivo prompting was effective for the third participant. Further, bids for joint attention did not generalize across novel items for any of the participants. Results are discussed in terms of the social relevance and validity of the use of video modeling to teach a complex social behavior such as bids for joint attention. Copyright © 2014 John Wiley & Sons, Ltd.

Joint attention can be described as the coordinated exchange of attention between two individuals and an object (Loveland & Landry, 1986) through the use of multiple topographies; some of which include shift in eye gaze, gestures (pointing or showing), vocal, or verbal communication using one or more words ('Look it's a ball' or 'Ball') or any combination of the text mentioned earlier. Adamson and Bakeman (1984) defined joint attention as an interaction that involves two people sharing attention with respect to an object or event, and monitoring each other's attention to that object or event. Joint attention is often discussed as one of the first forms of communication that emerges. Nonverbal joint attention behaviors such as eye contact and showing or pointing to objects emerge between 9 and 18 months in typically developing children (Adamson & Bakeman, 1984). Thus, joint attention skills are regarded as an important developmental milestone in terms of promoting language and social development (Wetherby, Prizant, & Schuler, 2000).

The ability to follow another person's focus of attention, as well the ability to direct another's focus of attention, allows children to establish a common topic with

*Correspondence to: Alison M. Betz, The Scott Center for Autism Treatment, Florida Institute of Technology, 150 W. University Blvd., Melbourne, FL, 32901 USA. E-mail: abetz@fit.edu

a communicative partner, and thus make sense of language. In addition to facilitating language development, joint attention also helps children understand how to act in a social world, interpret the perspectives of others, and participate in the back-and-forth nature of social exchanges (Schertz & Odom, 2004). Advanced levels of joint attention are characterized by the use of gestures such as pointing and reaching or showing a toy to a person, but topographies vary with the developmental level of the child. For example, younger children may use only eye contact, but older children tend to use a combination of conventional gestures that may include eye contact, pointing, and showing (Seibert, Hogan, & Mundy, 1984).

There are two forms of joint attention that may be taught together as one complete skill, or broken down into separate components. The first form is responding to bids for joint attention. This behavior may include shifting eye gaze from the conversational partner to the object of interest, emitting a vocal response to the bid for attention, or both. The second is initiating bids, for joint attention. This behavior typically includes getting someone's attention by emitting a verbal statement, pointing in the direction of an item of interest, showing the item to another individual, shifting eye gaze from the object of interest and the conversation partner, or some combination of these behaviors. An individual may initiate a bid for joint attention to gain access to the attention of another person, share information regarding a stimulus, which is either unique to the environment or well-known, or engage in social interaction (shared communication) with a conversational partner.

When analyzing a bid for joint attention using a verbal behavior approach, a bid for joint attention may function as a mand (i.e., request) for attention or information. For example, a child may point to an unknown or novel item in the environment and state 'Look at that!', thus making it likely that the conversation partner will respond by following the child's gesture and eye gaze shift, and vocally respond with a comment such as 'Wow. That's a big airplane!' This type of mand may be functioning as both a mand for the conversation partner to attend to what the child is attending to, and a mand for information regarding the item (e.g., name of the item).

When looking at the general topography of a bid for joint attention response, it may be difficult to determine whether it functions as a mand or a tact. In both cases, an individual may orient toward an item, point to the item, and make a vocal statement such as 'It's a (stimulus name).' What separates the mand for joint attention from the tact is the consequence. By definition, a tact is maintained by generalized reinforcement, which may include social interaction or feedback, whereas a mand for joint attention is reinforced specifically by access to social interaction, which typically is directed toward relevant stimulus in the environment. Thus, a critical component of a bid for joint attention, topographically, is the initiation of eye gaze shift from the target stimulus to the conversation partner and back. This eye gaze shift may, in fact, be the only variable that signals the conversation partner to respond socially toward the target stimulus.

Given the consequence that maintains an individual's bids for joint attention responses is social in nature, it is not surprising that this repertoire lacking in most children diagnosed with autism (MacDonald et al., 2006). The absence in the development of joint attention skills often observed in children with autism may be because of children with autism often not finding social interaction from others reinforcing. Joint attention has been linked to increased language development (Bono, Daley, & Sigman, 2004; Kasari, Freeman, & Paparella, 2006) and those who seek to engage in social interactions show an increase in the use of appropriate language. On the other hand, the inability to use language or engage in shared peer attention by a child with autism may lead to social isolation and a lack of understanding of the environment around them. Responding to and listening to others communicate can help to facilitate imitative communication and thus expand a child's language repertoire. Once an individual learns to respond to and initiate bids for joint attention, the level of communication and information about the surrounding environment that the individual is exposed to may be dramatically increased.

Numerous studies have analyzed different ways of teaching bids for joint attention including parent training (Jones & Feeley, 2007; Rocha, Schreibman, & Stahmer, 2007; Schertz & Odom, 2004), script fading (MacDuff, Ledo, McClannahan, & Krantz, 2007), discrete trial training (Kasari et al., 2006; Whalen & Schreibman, 2003), natural environment teaching (Koegel, Vernon, & Koegel, 2009; Whalen & Schreibman, 2003), and the use of pivotal response training (Pierce & Schreibman, 1995; Rocha et al., 2007; Vismara & Lyons, 2007). In many of these cases, authors also contrive motivation by incorporating highly preferred items into training as both antecedents and consequences.

Because of the complexity of teaching joint attention (both responding to and initiating bids), the skill is often broken down into multiple components with each being taught in succession. Pointing and/or orienting a person's body in the direction of an object in the immediate environment, verbal statements, and eye gaze shift may all be taught individually as a result. Teaching each component separately may make it more likely that the child will learn each component. In a recent review of the joint attention literature by Meindl and Cannella-Malone (2011), the authors reported that the best way to teach joint attention is by using simple and least restrictive procedures. The most common treatment packages to teach responding to and initiating bids for joint attention consisted of using graduated prompting (most-to-least intrusive or least-to-most intrusive), prompt fading, and included gestural, visual, and light physical guidance of the head and body in vivo or as the response occurs.

Although several researchers have demonstrated the positive effects of using graduated guidance to teach joint attention, there are also possible drawbacks to consider. For example, this teaching procedure may promote more rigid responding; especially when the components of a response chain such as joint attention are taught in isolation.

The brief pause that between the opportunity to respond and the prompt to respond when teaching each component of the joint attention skill in isolation or separately, may result in rigid or rote behaviors, or prompt dependence within the behavior chain. Results of studies that use scripts to teach bids for joint attention support this hypothesis by showing little to no deviations from scripted responses or the use of spontaneous initiations (e.g., Pollard, Betz, & Higbee, 2012).

In addition, more complex teaching procedures may promote inappropriate stimulus control. For example, Taylor and Hoch (2008) extended previous research on teaching bids for joint attention to children with autism using prompting (most-to-least prompting procedures) and differential reinforcement in the form of social praise or interaction. Results showed that more complex teaching procedures were needed to teaching some participants to initiate bids for joint attention. Specifically, for one participant, the researchers added textual prompts and tangible reinforcers to increase independent bids for joint attention. By including such supplemental procedures, it was unclear whether the targeted response was under appropriate stimulus control. Likely, the response was under the discriminative control of the textual prompt and was maintained by the tangible reinforcer, not the social interaction itself.

One procedure that removes the necessity of physical prompting within the behavior chain is video modeling. During video modeling, an individual watches a video clip of a target behavior being performed either by an adult or peers and then is given an opportunity to perform the target behavior with the use of prompts, which can be faded when needed (Charlop-Christy, Le, & Freeman, 2000). As a result, this type of teaching procedures may be beneficial to use when teaching both responding to and initiating bids for joint attention.

Due to its known advantages, video modeling has been demonstrated to be an effective intervention tool to teach a variety of social play skills (Charlop & Milstein, 1989; MacDonald, Clark, Garrigan, & Vangala, 2005; Paterson & Arco, 2007; Sancho, Sidener, Reeve, & Sidener, 2010), daily living skills (Murzynski & Bourret, 2007), social interactions (Bellini, Akullian, & Hopf, 2007; Charlop & Milstein, 1989; Wert & Neisworth, 2003), imitation (Cardon & Wilcox, 2011), and a variety of other behaviors including socially expressive behaviors (Charlop, Dennis, Carpenter, & Greenberg, 2010). Video modeling has been implemented extensively in schools to teach a variety of behaviors including on-task behavior, compliance to teacher requests, hand raising, and reduction of disruptive behavior (Hitchcock, Dowrich, & Prater, 2003).

Despite the positive effects demonstrated in the literature with other social behavior, to date, there are no published studies that examine the effects of using video modeling alone to teach initiating bids for joint attention. Given the success reported by researchers with teaching other complex social behaviors such as social interaction, play skills, and social initiations using video modeling, it may be hypothesized that video modeling procedures may be effective for teaching mands children with autism and

other intellectual disabilities to mand for joint attention as well. Thus, the purpose of this study was to examine the extent to which video modeling can be used to teach mands for joint attention to young children with autism or other developmental disabilities. Specifically, we examined the use of video modeling to teach bids for joint attention by having the children watch a video model of a peer emitting mands for joint attention toward an adult. In the video, the components of a complete mand for joint attention (i.e., pointing and/or orienting to the stimulus, emitting a vocal statement, and shifting eye gaze to and from the conversation partner and stimulus) were highlighted. If some or all of the components of the initiation did not emerge, we then examined the effects of in vivo prompting in addition to video modeling to teach the specific component response(s) that were not acquired with video modeling alone.

METHOD

Participants and Setting

Three participants, two boys and one girl, were included in the study. All participants had a diagnosis of autism and at the time of the study were receiving between 9–15 h of behavioral intervention services at a behavioral-based autism center. Prior to experimental sessions, participants' current social communication repertoire was assessed using the Early Social Communication Scales (ESCS), which specifically examines an individual's ability to initiate bids for joint attention and other behavioral requests. Results of the ESCS showed that none of the participants were able to independently emit a complete bid for joint attention.

Bryce was a 5-year-old boy with autism who demonstrated an intermediate speaker and listener repertoire. He scored a 39.5 on the Childhood Autism Rating Scale (CARS), which indicated he fell within the severely autistic range. He communicated primarily using 2–3 word sentences, was able to imitate peers and adults both in vocal communication and gross motor movements, was able to independently complete a sequence of five instructions, and physically approached others to initiate interactions. Testing on the ESCS indicated he was able to follow a point from an adult to an object and readily made eye contact with the therapist when his name was called. In addition, Bryce was also tested on his delayed imitation. He was able to imitate a variety of gross motor movements with a delay of 3 s.

Spencer was a 5-year-old boy with autism. His CARS score of 35 put him in the mild to moderately autistic range. Spencer's speaker and listener skills were advanced, as he often communicated using 4–5 word sentences, verbally interacted with peers when prompted, and imitated adults and peers. Testing on the ESCS indicated he demonstrated both skills of responding to bids for joint attention (able to follow a point,

follow an adult eye gaze). He also made eye contact with the therapist when name was called and upon the therapist giving instructions.

Alyssa was a 5-year-old girl with autism. Her CARS score of 31 indicated she fell within the mild to moderately autistic range. Alyssa was able to communicate using 4–5 word sentences, but did engage in echolalia as well. She demonstrated strengths in the areas most related to joint attention behaviors including compliance with adults, imitation, social interaction, and requesting. Further, she frequently requested using 10 or more words, and verbally interacted with peers when prompted. The ESCS indicated she was able to make eye contact with an adult and follow a point to an object.

Sessions took place at a university-based autism center that provided behavioral services in which the participants were currently enrolled as clients. All sessions were conducted in the hallway of the early intervention wing of the building and in the adjacent classrooms. Video models were shown in the participants' individual treatment rooms where they worked one on one with a trained therapist on skills such as communication, play, academic tasks, and daily living skills. Rooms were free of other individuals and distractions, and only contained relevant equipment to watch the video model (iPad®, table, and chairs). The therapist conducting sessions was present in the room with the participant, and the secondary data collector observed the presentation of the video model via a one-way mirror. Once the child watched the video, the secondary data collector moved to take position at the far end of the hallway before the participant and therapist left the treatment room. There was an approximate break of 10 s from the end of the video until the child and therapist entered the hallway. During all conditions, the hallway was free of other clients and staff at the autism center, with the exception of the data collector. In addition, all doors connecting to the hallway were closed to minimize distractions. The hallway contained multiple visual stimuli on the walls (e.g., decorations, bulletin boards, and pictures of clients), which remained in place throughout all sessions, and all participants had been exposed to such stimuli regularly.

Video Model and Materials

All objects used in this study were present during baseline and treatment sessions and were immediately removed once sessions concluded. Ten stimuli unique to the environment (e.g., inflatable palm tree, brightly colored pompoms, and construction cones) were strategically placed throughout the hallway and were present across all sessions. Of the 10 objects, five were identical to those used in the video model. The remaining five objects were used to test for generalization across stimuli and were not stimuli shown in the video model. The five generalization objects were randomly selected from an array of eight items and rotated across sessions, thus

not all generalization items were used every session. The generalization objects were placed before and after each target object from the video approximately 2 m apart from each other.

The video model that was used for all sessions and participants was 68 s and consisted of a 5-year-old girl and adult dyad demonstrating each component of a bid for joint attention. In the video, the child modeled bids for joint attention for five target items that were unique to find in a hallway of a learning environment to the adult. The child in the video emitted each individual component of a bid (i.e., pointing and/or orienting, emitting the vocal statement 'look,' along with a tact of the object, and eye gaze shift from the object to the therapist and back to the object). Following each bid for joint attention, the adult responded with a brief statement relevant to the target object (e.g., 'Wow, that tree is so cool!').

Response Definitions and Measures

The primary dependent variable for this study was independent bids for joint attention. An independent bid for joint attention was defined as emitting all of the following responses: (a) orienting and/or pointing to the targeted stimulus; (b) emitting a vocal statement pertaining to the target stimulus; and (c) shifting eye gaze from the target object to the conversation partner in the absence of any prompts. Each opportunity to initiate a bid for joint attention served as a trial. Opportunities were defined as walking toward or by one of the target or generalization stimuli placed throughout the hallway. Data were collected on a trial-by-trial basis and were reported as the percentage of trials in which an independent bid was emitted by dividing the number of independent bids by the total number of opportunities to emit a bid (i.e., 10) and multiplying by 100%. Further, data were collected on whether the participant independently emitted each component of the bid for joint attention to determine whether the participant acquired some, but not all, of the necessary components (e.g., the participant acquired the pointing response, but not the eye gaze or vocal response). The data for each component was also reported as a percentage score using the calculation described earlier. The succeeding text are operational definitions for each component that was measured.

Pointing and/or orienting

Pointing was defined as the extension of an isolated index finger in the direction of the target stimulus. *Orienting* was defined as a child turning the front of his or her body position including a head turn within 45 degrees to the direction of the stimulus. The child may or may not turn their body position back toward the adult. An independent occurrence was scored if the child demonstrated the behavior before moving

past the target stimulus. An error was recorded if the child passed the target stimulus without pointing and/or turning toward the stimulus.

Vocal statement

A *vocal statement* was defined as an utterance of a single word or phrase, or an approximation of the word or phrase that pertains to an object in the environment (MacDuff et al., 2007). The vocal statement had to be audible and clearly articulated from a distance of 1 m. The statement could either be scripted from the video (e.g., identical to the model in the video) or unscripted (e.g., different from the vocal statement modeled in the video). The scripted response from the video included the word 'look' plus the item name (e.g., 'Look, tree'). However, if the participant emitted the word 'look' alone or emitted the name of the object alone ('tree'), a vocal statement was scored as occurring. The vocal statement was scored as an independent occurrence if the participant emitted the statement within 2 s of approaching or orienting to the target stimulus. The vocal statement was scored as a nonoccurrence if the participant did not emit a statement or emitted the statement following 2 s of approaching or orienting to the target stimulus. The vocal statement was also scored as a nonoccurrence if the participant emitted a statement that did not pertain to the target stimulus (e.g., saying 'ball' in the presence of a puzzle).

Eye gaze shift

Eye gaze shift was defined as the movement of eyes from the conversation partner's face to the target stimulus. Eye gaze shift may include shifting gaze back to the conversation partner but was not required. To be scored as an independent occurrence, the eye gaze shift must occur simultaneously with the vocal statement or no more than 2 s following the vocal statement.

Data collection

Trial-by-trial data were collected on whether the participant emitted each component of a bid for joint attention by recording a plus or minus (+/−). At the end of each session, the data collector calculated the percentage of trials in which the participant independently emitted a complete bid for joint attention (i.e., independently emitted all components described earlier), as well as the percentage of independence for each component of the bid. A second independent observer collected data on the occurrence of each response described earlier using procedures identical to those described in the aforementioned text. The second observer stood at the end of the

hallway throughout the session. Trial-by-trial interobserver agreement (IOA) was calculated by comparing the primary data collector's data to the data of an independent observer for each of the three target behaviors (pointing/orienting, vocal statement, and eye gaze shift) for every trial. IOA scores were calculated by dividing the total number of agreements by the total number of opportunities to respond and multiplying by 100 to yield a percentage score. IOA was collected for 82%, 64%, and 44% of sessions across all conditions for Bryce, Spencer, and Alyssa, respectively. Mean IOA across all conditions for Spencer was 100%. Mean IOA across all conditions was 98.9% (range, 90 to 100%) for Bryce and 98.7% (range, 90 to 100%) for Alyssa.

Experimental Design and Procedures

A delayed multiple baseline across participants design was used to evaluate the effects of video modeling on the acquisition of components of, and completed bids for, joint attention. We chose this design specifically to attempt to control for habituation to the objects in the environment. All participants were exposed to baseline and video modeling conditions, and Alyssa was exposed to a video modeling + in vivo training condition.

Baseline

All sessions consisted of 10 trials and were approximately 5 min in duration (session duration varied across participants and sessions). One to three sessions were conducted per day with at least 30 min, separating each session. During each condition, the session started when the therapist gave the instruction, 'Let's take a walk' or 'Walk this way'. Following the instruction, the therapist allowed the participant to walk through the hallway containing the target stimuli and remained within arm's reach, either behind or beside, of the participant at all times. If the participant walked a distance of greater than 1 m past the stimuli without responding, no programmed consequences were provided and that trial ended (e.g., the opportunity to respond passed). If the participant walked by stimuli to a distance of greater than 1 m, but then attempted to return to the stimuli, the attempt was blocked and the participant lost the opportunity to engage in the bid. All problem behavior was ignored and the participant was redirected to continue to walk through to the end of the hallway.

All consequences for partial or complete bids for joint attention remained constant across training stimuli (those shown in the video model) and generalization stimuli. Differential reinforcement in the form of a verbal statement was provided to the child if he or she completed a partial bid for joint attention (i.e., emitted at least one of the three required responses). The verbal statement was provided in a neutral tone of voice and was limited to one or two words (e.g., 'yes') and a head nod. After each

independent and complete bid for joint attention, the therapist responded with an enthusiastic statement that included the stimulus name (e.g., 'Wow, that palm tree, is cool!'). The therapist also shifted his or her gaze between the object and child in response to the child's gaze alternation and pointing responses. For items that the child walked past a minus (–) was scored for that trial. The therapist responded to any further comments regarding the target items that the participant emitted.

Video modeling

Video modeling sessions were conducted identical to those described in baseline with one exception. Prior to each session, the participant watched the video model. During this condition, the participant remained in his/her individual treatment room, free of distractions and was given the instruction, 'It's time to watch a movie' or 'We are going to watch a movie now' and the therapist started the video. The participant's behavior while watching the video was monitored to ensure that he or she was attending to the video model. Specifically, the therapist observed the direction in which the participant's eyes were focused. If the participant's eyes were not directed toward the screen in which the video was being played for a duration great than 3 s, the therapist verbally prompted the participant to attend to the video (e.g., 'watch the video'). The therapist tracked the total duration of attending to the video using a stopwatch. If the total duration of attending was less than 48 s (e.g., the participant's eyes were not oriented toward the screen for more than 20 s of the 68-s video), the experimental session was terminated for that day. Once the participant watched the full video, the experimental session began and sessions were conducted in the same manner described in baseline. It should be noted that termination of a session as a result of not attending to the video did not occur for any of the participants.

Video modeling + in vivo prompting (VM + IND)

For Alyssa, in vivo prompts were required for the eye gaze shift component. In this condition, sessions were identical to those described earlier; however, the therapist prompted eye gaze shift from the target item to the therapist using a gentle chin prompt. Specifically, if Alyssa did not shift her eye gaze from the target object to the therapist within 3 s of completing the other two components (point/orienting and vocal), the therapist gently placed her hand under Alyssa's chin and guided her face toward the direction of the therapist. Once the participant successfully emitted the eye gaze shift, the therapist provided enthusiastic social praise ('Wow! Yeah, good looking, Look at all the colors of the rainbow'), and moved to the next object where the procedure was repeated. Eye gaze was prompted only for target stimuli.

Treatment integrity was obtained for 42%, 43%, and 47%, of sessions across all conditions for Bryce, Spencer, and Alyssa, respectively. Mean treatment integrity across all conditions was 100% for all participants.

RESULTS

Figure 1 shows the percentage of opportunities with independent and complete bids for joint attention emitted for all participants. During baseline, Bryce (top panel) did not emit any complete independent bids for joint attention toward target or generalization items. Once video modeling was introduced, Bryce’s mean percentage of complete independent bids was 73% (range, 20 to 100%) for target objects. The low percentage scores reflected in the range were due to the initial low levels of responding. Bryce initially only began responding with a complete and independent bid toward the first target stimulus he approached when video modeling was introduced (sessions 4–6). However, this was followed by a steady increase in complete

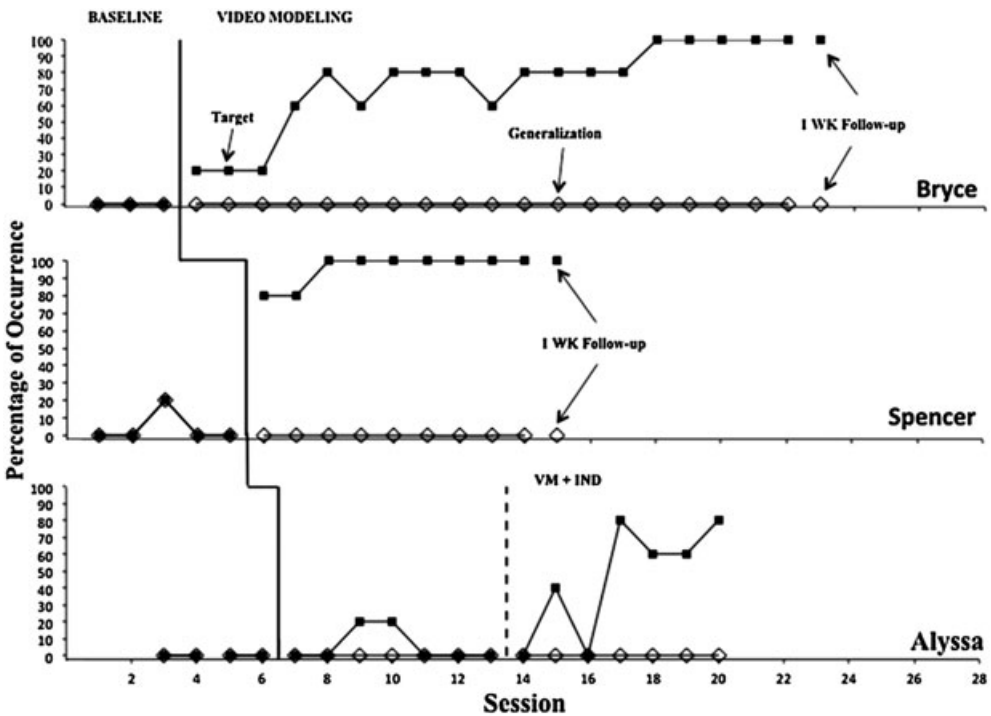


Figure 1. Percentage of opportunities in which a complete independent bid for joint attention occurred for Bryce (top panel), Spencer (second panel), and Alyssa (bottom panel).

independent bids, with correct responding reaching 100% after 14 sessions. Bryce's correct responding maintained at 100% for the last five sessions and at a 1-week follow up. Bryce never emitted a complete independent bid for joint attention toward the generalization stimuli.

Bryce's percentage of correct responding for each individual component (top panel of Figure 2) showed that he emitted some components of joint attention independently during baseline. Specifically, the mean percentage of independent pointing/orienting during baseline was 46% (range, 40 to 60%); however, Bryce never emitted a vocal statement or engaged in shifts of eye gaze during baseline sessions. After implementing the video modeling condition, the percentage correct for all components increased for pointing/orienting ($M=98\%$; range, 80 to 100%) vocal statements ($M=96\%$; range, 60 to 100%), and eye gaze shift ($M=73\%$; range, 20 to 100%).

Spencer (second panel of Figure 1) emitted one complete independent bid for joint attention toward a target stimulus and one generalization stimulus in one session during baseline. After video modeling was implemented for Spencer, there was an

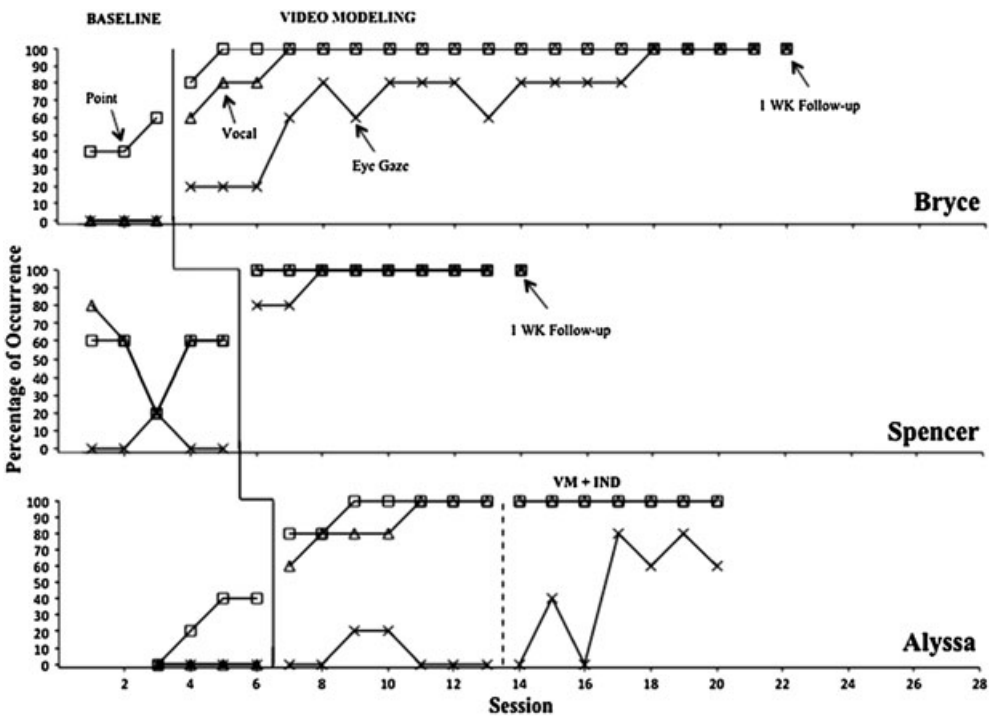


Figure 2. Percentage of opportunities in which each independent component of a mand for joint attention occurred for Bryce (top panel), Spencer (second panel), and Alyssa (bottom panel).

immediate increase in complete independent bids to 80% of opportunities for target objects. After only three sessions, Spencer was emitting a complete independent bid for joint attention across all five target objects from the video. During the video modeling condition the mean percentage of Spencer's complete independent bids for joint attention was 96% (range, 80 to 100%) for target stimuli. Complete independent bids for generalization objects remained at 0% across all sessions once video modeling was introduced. For Spencer's individual component data (second panel of Figure 2), the mean percentage for pointing/orienting was 55% (range, 20 to 80%); 60% (range, 20 to 80%) for vocal statements; and 4% (range, 0 to 20%) for eye gaze shift during baseline. After implementing the video modeling condition, the mean percentage increased to 100% for pointing/orienting, and vocal statements, and 96% (range, 80 to 100%) for eye gaze shift.

Alyssa did not emit any complete independent bids for joint attention during baseline for the target or generalization stimuli (third panel of Figure 1). Once video modeling was implemented for Alyssa, we saw an increase in complete independent bids for target objects to 20% after three sessions; however, her responding did not maintain and returned to baseline levels for the next three sessions ($M=6\%$; range, 0 to 20%). Alyssa's individual component data (third panel of Figure 2) showed that during baseline, she independently emitted the pointing/orienting response to some stimuli ($M=25\%$; range, 0 to 40%) but never emitted vocal statements or eye gaze shifts. During the video modeling condition, the mean percentage increased to 94% (range, 80 to 100%) for pointing/orienting; 86% (range, 60 to 100%) for vocal statements; and 6% (range, 0 to 20%) for eye gaze shift.

After video modeling was implemented, Alyssa demonstrated an immediate increase from baseline levels of 25% to 94% for pointing/orienting and from 0% to 86% for vocal statements after six sessions. However, her eye gaze shift only slightly increased from 0% to 6% once video modeling was introduced. Once responding reached 100% across three consecutive sessions for both pointing/orienting and making a vocal statement toward target objects, while eye gaze shifts remained at low levels, we implemented in vivo prompting (VM+IND) for the eye gaze component. During this condition, the mean percentage of independent eye gaze shifts increased to 46% (range, 0 to 80%), whereas her pointing/orienting and vocal responses maintained at 100%. Thus, adding the eye gaze prompt was sufficient to increase the percentage of complete independent bids for Alyssa ($M=46\%$; range, 0 to 80%). Further, for two of the seven sessions conducted following in vivo prompting, Alyssa emitted the eye gaze shift for 80% of opportunities. However, she did not emit any of the individual components or a complete independent bid for joint attention toward any of the generalization stimuli.

DISCUSSION

The results of this study suggest that video modeling alone was effective in increasing complete and independent bids for joint attention for two of the three participants and was effective at increasing some of the components of the target response for the third participant. The treatment package (video modeling in combination with in vivo prompts) was effective at increasing complete and independent bids for joint attention for Alyssa. This study supports the findings of previous research on teaching bids for joint attention in the MacDonald et al. (2006) study, which indicated that some children with autism engage in some of the components of joint attention without specific instruction, and the more complex components like eye gaze shift may require more intense instruction.

This study contributes to the current literature on teaching mands for joint attention in several ways. First, to our knowledge, this study is one of the first that successfully used video modeling alone to teach a complex interaction such as joint attention. Further, this study was one of the few that used social reinforcement alone to teach bids for joint attention and did not use access to tangible or preferred stimuli as reinforcement for emitting bids for joint attention. Within the current literature, only two studies (Naoi, Tsuchiya, Yamamoto, & Nakamura, 2008; Taylor & Hoch, 2008) exclusively used adult attention as a consequence for joint attention initiations (with the exception of one participant in the Taylor & Hoch, 2008 study). Our study replicated the findings of Taylor and Hoch (2008), in that procedures used to reinforce complete bids (e.g., commenting on the unusual item with socially acceptable praise only) were successful for some, but not all of the participants. This finding is particularly important, as it demonstrates that for some children, social attention alone may be sufficient to teach or increase independent bids for joint attention. In addition, using social interaction or attention alone as the consequence provides a clear demonstration that the initiation was not simple and mand for access to a specific item.

Second, few studies have investigated the effects of video modeling without the use of additional intervention strategies. This study did not prompt bids for joint attention during baseline or the video model condition. Further, the therapist did not direct the child toward the objects while walking in the hallway, nor did they provide any physical or verbal prompting during the video modeling condition. This allowed us to evaluate video modeling as an intervention alone to determine its effects on teaching mands for joint attention. In this study, two of the three participants learned to emit independent bids for joint attention using video modeling alone, which serves as preliminary data that show the utility, practicality, and social significance of video modeling. Future researchers, of course, should replicate this study with additional individuals, and perhaps begin to evaluate what individual characteristics or repertoires may contribute to the success of the intervention.

Another notable extension of the current literature on teaching bids for joint attention resulting from the study is the inclusion of individual component data to determine whether the complete response was truly functioning as a bid for joint attention. By measuring each specific component of the joint attention response, we were able to determine whether the eye gaze shift, which may be considered the most critical component of a bid for joint attention, was present. For two of the three participants (Spencer and Bryce), we observed acquisition and maintenance of all three individual components with video modeling alone. This suggests that for these participants, the response was likely functioning as a bid for joint attention, and not an example of tacting the item. On the other hand, video modeling alone was not sufficient at teaching a bid for joint attention for Alyssa. It may be case, that for Alyssa, the video model was an ineffective intervention simply because the eye gaze shift modeled in the video was not salient enough for her to observe in the video. Thus, future research may want to use more sophisticated technology (e.g., slow motion or other enhancement features) when making video models that could increase the saliency of each component of the bid for joint attention.

Although the results of this study are promising, it is not without limitations. First, we did not include generalization probes across additional settings. All sessions were conducted with the objects placed in the same environment that the child was exposed to during training (i.e., the video modeling condition). In addition, we did not include procedures to increase the generalization across stimuli. Future researchers should extend this research by evaluating methods in which video models can be modified to teach a more generalized repertoire of bids for joint attention.

Third, we did not account for possible habituation effects. Thus, the low levels of independent bids for joint attention observed with Alyssa may have been due to the participants losing interest in the items used to evoke the bids for joint attention. Quite often, people engage in joint attention because they see something that is of interest to them. Once an individual has come into contact with the object one or more times, the object may become less interesting, and the motivation to acquire social attention from pointing out the object may dissipate. We may habituate to its presence, and thus refrain from emitting a bid for joint attention, specifically, targeting that object. Future researchers should consider conducting follow up probes with baited items in new settings and the possibility of multiple videos demonstrating the behavior in a variety of settings with objects throughout.

Although these data are preliminary, the results of this study suggest that video modeling may be an effective tool to teach bids for joint attention, which has practical implications because of its ability to remove the therapist from prompting the individual. Because the videos can often be between 1 and 3 min in length, they may provide effective options for educators to design interventions to teach complex skills in environments where time may be limited. In addition, it can be used for a variety

of skills, may be effective for all ages, and can be implemented in a variety of settings. Regardless of the intervention technology implemented, future research must continue to evaluate procedures that enhance generalization of bids for joint attention to the natural environment.

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